

Part VI — Heat, Temperature, and Thermodynamics · Chapter 33

Laws of Thermodynamics

The universal rules governing energy, heat, work, and the direction of natural processes

Every machine ever built, every engine ever conceived, every living cell on Earth operates under the same inescapable constraints. You can transform energy from one form to another, but you cannot create it from nothing and you cannot destroy it. You can move heat from a hot body to a cold one, but you cannot spontaneously move it from cold to hot without doing work. You can organise disordered matter into structure, but you always increase the disorder somewhere else in the process. These are not technical limitations to be engineered around — they are the laws of thermodynamics, universal constraints on what physical processes can and cannot do. This chapter presents these laws clearly, explains the physical reasoning behind each one, and shows how they govern everything from steam engines to living organisms to the ultimate fate of the universe.

33.1 Why Thermodynamics Has “Laws”

Physics has many “laws” — Newton’s Laws, the Law of Conservation of Momentum, Snell’s Law — but the laws of thermodynamics have a special status. They are more fundamental in a specific sense: they constrain entire classes of processes, not just individual interactions. No violation of the thermodynamic laws has ever been observed in any macroscopic system, despite centuries of attempts. Their status is comparable to logical constraints rather than empirical regularities.

The laws of thermodynamics emerged from 19th-century attempts to understand and improve steam engines. Sadi Carnot, Rudolf Clausius, William Thomson (Lord Kelvin), and James Joule derived universal principles that turned out to apply to every physical system. What began as engineering thermodynamics became one of the most profound theories in all of science.

There are four laws, numbered somewhat counterintuitively: the Zeroth, First, Second, and Third. The Zeroth defines temperature and equilibrium. The First is conservation of energy. The Second is the most famous — it introduces entropy and the arrow of time. The Third sets an absolute limit on how cold anything can be. Together they form a complete and closed framework for understanding all thermal processes.

Historical Note: The term “thermodynamics” was first used by Lord Kelvin in 1849. The Second Law was formulated almost simultaneously by Clausius (“heat cannot pass from a colder to a warmer body on its own”) and Kelvin (“a temperature difference cannot be produced by a cyclic process that does no work”). These two statements are logically equivalent, though they look superficially different.

33.2 Thermodynamic Systems and Boundaries

A thermodynamic system is any defined region of matter or space under study. The rest of the universe — everything outside the system — is the surroundings or environment. The boundary separates them. Specifying the system and its boundary is the first step in any thermodynamic analysis.

Systems are classified by what can cross their boundaries. An open system can exchange both matter and energy with the surroundings (an open pot of boiling water). A closed system can exchange energy (as heat or work) but not matter (a sealed gas cylinder). An isolated system exchanges neither matter nor energy with the surroundings (an ideal thermos bottle). Most real systems are approximately closed or open; truly isolated systems are idealisations useful for analysis.

System Type	Matter Exchange?	Energy Exchange?	Example
Open	Yes	Yes	Boiling pot, living cell, ocean
Closed	No	Yes	Sealed cylinder, steam engine piston
Isolated	No	No	Ideal thermos, theoretical universe

Table 33.1 *Classification of thermodynamic systems by boundary permeability.*

33.3 State Variables and System States

The thermodynamic state of a system is completely described by its state variables: macroscopic properties that have well-defined values when the system is in equilibrium. For a simple gas, the state variables are pressure P , volume V , temperature T , and amount n . Any one of these can be calculated from the others using the equation of state (ideal gas law for an ideal gas).

A process is a change from one state to another. Processes are classified by which

variable is held constant: isothermal (constant T), isobaric (constant P), isochoric or isovolumetric (constant V), and adiabatic (no heat exchange, $Q = 0$). Each type has characteristic equations derived from the First Law.

A thermodynamic cycle is a sequence of processes that returns the system to its original state. The net change in all state variables over a complete cycle is zero — the system is back where it started. This means the net change in internal energy is also zero ($\Delta U_{\text{cycle}} = 0$), which has profound implications for heat engines (Chapter 34).

Not all quantities are state variables. Heat Q and work W are not state variables — they are path functions: their values depend on the process taken, not just the initial and final states. This distinction is critical to the First Law.

33.4 The Zeroth Law of Thermodynamics

The Zeroth Law was introduced in Chapter 29, but its deep significance in the framework of thermodynamics deserves fuller discussion here. The Zeroth Law states: if system A is in thermal equilibrium with system C, and system B is in thermal equilibrium with system C, then A and B are in thermal equilibrium with each other.

This law is the logical foundation for the concept of temperature as a universal measurable quantity. It says that “being at the same temperature” is a transitive equivalence relation — it partitions all physical systems into equivalence classes, each defined by a common temperature. Without the Zeroth Law, temperature could not be consistently defined or measured by thermometers.

The Zeroth Law also implies that when two systems are brought into thermal contact, they will eventually reach a common equilibrium temperature — and that temperature is measurable by any third system (thermometer) in equilibrium with them. This is such a foundational concept that it was given the number zero to emphasize it precedes everything else.

33.5 The First Law of Thermodynamics

The First Law of Thermodynamics is the law of conservation of energy applied to thermodynamic systems. It states that the change in internal energy of a system equals the heat added to the system minus the work done by the system:

$$\Delta U = Q - W$$

First Law of Thermodynamics. ΔU = change in internal energy (J); Q = heat added to system (J, positive in); W = work done by system (J, positive out). Note: some textbooks define W as work

done ON the system, giving $\Delta U = Q + W$. Always check the sign convention.

This equation says that the internal energy of a system can be changed in exactly two ways: by adding or removing heat (Q), and by doing work on or by the system (W). There is no other way. Energy cannot be created or destroyed; it can only be transferred or transformed.

The First Law also says that internal energy U is a state variable: ΔU depends only on the initial and final states, not on the path taken. Even though Q and W individually depend on the path, their combination $Q - W$ does not. This is why U is a state function and Q , W are not.

33.6 Internal Energy, Heat, and Work

Internal energy U is the total microscopic energy of the system — kinetic energy of molecules plus potential energy of intermolecular forces. For an ideal monatomic gas, $U = (3/2)nRT$. For real substances, U includes additional terms for rotational, vibrational, and potential energy.

Heat Q is thermal energy crossing the system boundary due to temperature difference. It is positive when energy flows into the system (from a hotter reservoir) and negative when energy flows out (to a cooler reservoir). Heat is a process, not a property.

Work W in thermodynamics refers specifically to mechanical work done by the system on its surroundings (or vice versa). For a gas, the most common form of work is expansion work (pressure-volume work): when a gas expands against an external pressure P_{ext} by a volume change ΔV , the work done by the gas is $W = P_{\text{ext}} \Delta V$ (for constant external pressure). In general, W is the area under the P - V curve.

$$W = P\Delta V \quad (\text{constant pressure process})$$

Work done by a gas at constant pressure. Positive when the gas expands ($\Delta V > 0$); negative when compressed ($\Delta V < 0$, work done on the gas).

Example — First Law calculation A gas absorbs 500 J of heat and does 200 J of work on a piston. $\Delta U = Q - W = 500 - 200 = 300$ J. The internal energy of the gas increases by 300 J.

Example — Isochoric process A gas in a rigid container (V constant, $\Delta V = 0$, so $W = 0$) absorbs 400 J of heat. $\Delta U = Q - W = 400 - 0 = 400$ J. All heat goes directly into increasing internal energy.

33.7 Energy Conservation in Thermal Processes

The First Law applied to specific process types gives useful results.

Isothermal process (T constant): for an ideal gas, U depends only on T , so $\Delta U = 0$. Therefore $Q = W$: all heat added equals work done by the gas. (A gas expanding isothermally at constant temperature absorbs heat from a reservoir and converts it entirely to work — but this requires maintaining exact thermal contact with a reservoir, which is an idealization.)

Adiabatic process ($Q = 0$): no heat exchange with surroundings. $\Delta U = -W$: any work done by the gas reduces its internal energy (and therefore its temperature). This is why a gas cools when it rapidly expands (adiabatic expansion). Diesel engines exploit this: fuel-air mixture is compressed adiabatically until the temperature rise ignites the fuel, without a spark plug.

Isochoric process (V constant, $W = 0$): $\Delta U = Q$. All heat added increases internal energy. This is the case for a gas in a rigid sealed container.

Isobaric process (P constant): $\Delta U = Q - P\Delta V$. Both heat and work contribute. Boiling water at constant atmospheric pressure is an isobaric process.

Process	Constraint	Simplification	First Law Result
Isothermal	T constant	$\Delta U = 0$ (ideal gas)	$Q = W$
Adiabatic	$Q = 0$	No heat exchange	$\Delta U = -W$; gas cools if it expands
Isochoric	V constant	$W = 0$	$\Delta U = Q$
Isobaric	P constant	$W = P\Delta V$	$\Delta U = Q - P\Delta V$

Table 33.2 First Law applied to four fundamental process types.

33.8 The Second Law of Thermodynamics

The First Law tells us that energy is conserved in any process. But it says nothing about which processes actually occur. Energy conservation is compatible with a coffee cup spontaneously getting hotter by absorbing heat from cooler air — energy would still be conserved, just flowing in the unusual direction. Yet this never happens. The Second Law explains why.

The Second Law has several equivalent formulations. The Clausius statement: heat cannot flow spontaneously from a colder body to a hotter body. The Kelvin-Planck statement: it is impossible for any cyclic process to convert heat from a single reservoir

entirely into work with no other effect. These statements are logically equivalent — a violation of one implies a violation of the other.

The Second Law is a statement about the direction of spontaneous processes. Processes go in the direction that increases the total entropy of the universe. This gives thermodynamics its “arrow of time”: the laws of mechanics are time-reversible, but the Second Law is not. You can always tell which direction time runs by watching entropy.

Practical consequences: no heat engine can be 100% efficient. No refrigerator works without doing work. Disorder (entropy) always tends to increase. Organised structures require energy input to maintain. All these follow from the Second Law.

“The second law of thermodynamics is the only physical law that distinguishes the past from the future. All other laws are time-symmetric. Entropy is the arrow of time.” — Arthur Eddington (paraphrased)

33.9 Entropy (Conceptual Introduction)

Entropy (symbol S) is the thermodynamic state variable that quantifies the disorder or randomness of a system — more precisely, the number of microscopic configurations (microstates) consistent with the macroscopic state of the system. The more microstates available, the higher the entropy.

Intuitively: a gas compressed into a corner of a room has fewer available positions for its molecules (fewer microstates) than the same gas spread throughout the room. Ice has fewer microstates than liquid water at the same temperature (the molecules are locked in a crystal lattice). A perfectly ordered deck of cards has far fewer arrangements than a shuffled deck.

The statistical definition (due to Boltzmann): $S = k_B \ln(\Omega)$, where Ω is the number of microstates and k_B is Boltzmann’s constant. This equation, inscribed on Boltzmann’s tombstone, connects the macroscopic concept of entropy to the microscopic counting of particle configurations. It is one of the most profound equations in all of physics.

$$S = k_B \ln \Omega$$

Boltzmann’s entropy formula. S = entropy (J/K); $k_B = 1.381 \times 10^{-23}$ J/K; Ω = number of microstates. More microstates $\rightarrow$ higher entropy.

The thermodynamic (Clausius) definition of entropy change is more operational: for a reversible process at temperature T , the change in entropy of the system is $\Delta S = Q_{\text{rev}}/T$. Heat flows naturally from hot to cold because this transfer increases the total

entropy of the universe: the cold body gains more entropy (Q/T_{cold}) than the hot body loses (Q/T_{hot}), since $T_{\text{cold}} < T_{\text{hot}}$ and Q is the same.

33.10 Why Heat Flows Spontaneously in One Direction

The Second Law's directional constraint — heat flows from hot to cold — has a profound statistical explanation. When heat ΔQ flows from a hot body at T_H to a cold body at T_C :

The hot body loses entropy: $\Delta S_H = -\Delta Q/T_H$ (negative, because energy leaves).

The cold body gains entropy: $\Delta S_C = +\Delta Q/T_C$ (positive, because energy enters).

Net change in total entropy of the universe:

$$\Delta S_{\text{universe}} = \Delta Q/T_C - \Delta Q/T_H = \Delta Q(1/T_C - 1/T_H)$$

Net entropy change when heat flows from T_H to T_C . Since $T_C < T_H$, the quantity $(1/T_C - 1/T_H) > 0$, so $\Delta S_{\text{universe}} > 0$. Entropy increases.

Since $T_C < T_H$, the gain in entropy of the cold body exceeds the loss of entropy from the hot body. The total entropy of the universe increases. The reverse process — heat flowing from cold to hot — would decrease the total entropy of the universe ($\Delta S_{\text{universe}} < 0$), which never occurs spontaneously. This is not an empirical observation waiting to be overturned; it follows directly from the statistical definition of entropy and the enormous number of particles involved.

For a reversible process: $\Delta S_{\text{universe}} = 0$ (entropy is unchanged, no direction is preferred). For any irreversible (real) process: $\Delta S_{\text{universe}} > 0$. This is the Second Law stated as an entropy principle: the entropy of the universe never decreases.

33.11 The Third Law of Thermodynamics

The Third Law of Thermodynamics states: the entropy of a perfectly crystalline substance at absolute zero (0 K) is zero. Equivalently: it is impossible to reduce the temperature of any system to absolute zero in a finite number of steps.

The first statement ($S = 0$ at $T = 0$) follows from Boltzmann's entropy formula: a perfect crystal at 0 K has exactly one microstate (every atom in its lowest energy position, perfectly ordered). $\Omega = 1$, so $S = k_B \ln(1) = 0$. Any imperfection or residual disorder means $\Omega > 1$ and $S > 0$ even at 0 K (residual entropy).

The second statement (impossibility of reaching absolute zero) is its practical consequence. To cool a system, you must transfer its entropy to the surroundings. As T

→ 0, the system's heat capacity approaches zero, and extracting any remaining heat becomes progressively harder. The cooling process asymptotically approaches but never reaches 0 K. Researchers have achieved temperatures of 100 pK (10^{-10} K) in laboratory settings, but 0 K itself remains forever just out of reach.

The Third Law establishes entropy as a quantity with an absolute scale, not just differences. It allows the calculation of absolute entropies, which is important in chemistry (standard entropies of substances, used in calculating whether reactions will proceed spontaneously).

33.12 Limits on Perfect Efficiency and Perfect Cooling

The Second Law imposes absolute upper limits on the efficiency of any heat engine and absolute lower limits on the work required by any refrigerator. These limits are independent of the design or materials of the machine — they follow from the impossibility of decreasing the total entropy of the universe.

The maximum efficiency of any heat engine operating between a hot reservoir at T_H and a cold reservoir at T_C (both in kelvins) is the Carnot efficiency:

$$\eta_{\text{Carnot}} = 1 - T_C/T_H$$

Carnot efficiency: the maximum possible efficiency for any heat engine operating between T_H and T_C . No engine can exceed this; only a perfectly reversible (Carnot) engine achieves it.

The Carnot efficiency is a direct consequence of the Second Law. To achieve 100% efficiency ($\eta = 1$), you would need $T_C = 0$ — an absolute zero cold reservoir. But the Third Law says you cannot reach 0 K. So no engine can be 100% efficient.

For refrigeration: the Second Law requires that work must be done to move heat from a cold reservoir to a hot one (against the natural direction). The minimum work required is also determined by the Carnot relationship. A perfectly efficient refrigerator (Carnot refrigerator) provides maximum cooling per unit of work, but real refrigerators always require more.

Example — Carnot efficiency example A steam turbine receives steam at $T_H = 500^\circ\text{C}$ (773 K) and exhausts to $T_C = 30^\circ\text{C}$ (303 K). Maximum (Carnot) efficiency: $\eta = 1 - 303/773 = 1 - 0.392 = 0.608 = 60.8\%$. No steam turbine, however perfectly designed, can exceed 60.8% efficiency between these temperatures.

33.13 Common Misconceptions About Thermodynamics

- Confusing the First and Second Laws. The First Law says energy is conserved. The Second Law says some conversions are forbidden even when energy would be conserved. A coffee cup cooling air to heat itself would satisfy the First Law but violate the Second.
- Thinking entropy always increases locally. Entropy can decrease locally (water freezing, organisms growing, crystals forming) as long as it increases by at least as much in the surroundings. The Second Law applies to the total entropy of the universe, not any one system.
- Believing evolution violates the Second Law. Organisms are open systems that export entropy to their surroundings (as waste heat and excretion) while building local order. The total entropy of the Earth-plus-Sun system increases. The Second Law is not violated by any biological process.
- Confusing the work W in the First Law with mechanical work outside thermodynamics. In thermodynamics, W specifically refers to energy crossing the system boundary as organised (non-heat) energy. It includes mechanical work ($P\Delta V$), electrical work, and other forms. Always identify the system boundary first.
- Thinking reversible processes are just slow processes. A reversible process is an idealised limit in which the system is always in equilibrium and no entropy is produced. Real processes are always irreversible to some degree. Reversible processes are approached, never achieved, no matter how slowly you go.
- Believing a perpetual motion machine of the second kind is merely very difficult to build. A PMM2 (a device that converts heat from a single reservoir entirely into work with no other effect) would violate the Second Law, not just be hard to engineer. Its impossibility is a fundamental law of nature.

Common Mistake: The First Law forbids creating or destroying energy. The Second Law additionally forbids certain transformations that conserve energy. Both laws are absolute and have never been violated in any macroscopic system in recorded history.

Chapter Summary

The laws of thermodynamics constrain all thermal processes. They are universal, inviolable, and govern everything from engines to living organisms to the cosmos.

- Zeroth Law: thermal equilibrium is transitive; temperature is well-defined and measurable.
- First Law: $\Delta U = Q - W$. Energy is conserved. Internal energy U is a state function; Q and W are path functions.

- Second Law: entropy of the universe never decreases. Heat flows spontaneously from hot to cold. No heat engine is 100% efficient. Maximum efficiency: $\eta_{\text{Carnot}} = 1 - T_{\text{C}}/T_{\text{H}}$.
- Third Law: entropy of a perfect crystal at 0 K is zero. Absolute zero cannot be reached in a finite number of steps.
- Entropy $S = k_{\text{B}} \ln \Omega$ (Boltzmann); $\Delta S = Q_{\text{rev}}/T$ (Clausius). Entropy measures microscopic disorder.
- Process types: isothermal ($\Delta U = 0$), adiabatic ($Q = 0$), isochoric ($W = 0$), isobaric ($W = P\Delta V$).

Key Terms

thermodynamic system: The defined region of matter or space under study; the rest of the universe is the surroundings.

open system: A system that can exchange both matter and energy with its surroundings.

closed system: A system that can exchange energy but not matter with its surroundings.

isolated system: A system that exchanges neither matter nor energy with its surroundings.

state variable: A macroscopic property (P, V, T, U, S) with a unique value in each equilibrium state; independent of the path taken to reach that state.

path function: A quantity (Q, W) whose value depends on the process taken, not just the initial and final states.

thermodynamic cycle: A sequence of processes that returns a system to its original state; net $\Delta U = 0$ over one cycle.

Zeroth Law of Thermodynamics: If A is in equilibrium with C, and B is in equilibrium with C, then A is in equilibrium with B; defines temperature as transitive.

First Law of Thermodynamics: $\Delta U = Q - W$; energy is conserved; the change in internal energy equals heat added minus work done by the system.

Second Law of Thermodynamics: The entropy of the universe never decreases; heat flows spontaneously from hot to cold; no cyclic engine converts heat entirely to work.

Third Law of Thermodynamics: The entropy of a perfect crystal at absolute zero is

zero; absolute zero cannot be reached in a finite number of steps.

entropy (S): The thermodynamic state variable measuring microscopic disorder; $S = k_B \ln \Omega$; increases in every irreversible process.

Carnot efficiency (η_{Carnot}): $1 - T_C/T_H$; the maximum theoretical efficiency of any heat engine operating between T_H and T_C .

reversible process: An idealised process in which the system is always in equilibrium and entropy is not produced; the theoretical limit of efficiency.

irreversible process: Any real process that produces entropy; cannot be exactly undone without leaving a net change in the universe.

isothermal: Process at constant temperature; for ideal gas, $\Delta U = 0$.

adiabatic: Process with no heat exchange ($Q = 0$); $\Delta U = -W$.

isochoric: Process at constant volume; $W = 0$; $\Delta U = Q$.

isobaric: Process at constant pressure; $W = P\Delta V$.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain the difference between a closed and an isolated system. Give one example of each.
2. Why is it said that heat and work are "path functions" while internal energy is a "state function"? Explain what this means physically.
3. A gas in a rigid container absorbs 500 J of heat. How much does its internal energy change? How much work does it do? Use the First Law explicitly.
4. Explain, using the concept of entropy, why heat flows spontaneously from hot to cold but never from cold to hot.
5. A student claims: "Water freezing decreases entropy, so it violates the Second Law." Explain the error. What must be true of the surroundings for freezing to occur?
6. What is the Carnot efficiency? Why can no real engine achieve 100% thermal efficiency? Use both the Second and Third Laws in your answer.
7. A gas expands adiabatically against a piston and does work on the piston. Does the gas temperature rise or fall? Apply the First Law to explain.
8. Explain the Third Law in both its entropy and temperature formulations. How are the two formulations connected?

Thermodynamic System Analysis

9. Classify each system and identify possible energy/matter exchanges: (a) the inside of a sealed car on a hot day, (b) a growing tree, (c) the Sun, (d) an ideal thermos of soup, (e) a running car engine.
 10. For each process, identify which type (isothermal, adiabatic, isochoric, or isobaric) best describes it: (a) air compressor pumping so fast there is no time for heat exchange; (b) a gas slowly expanding in a cylinder held at 0°C in ice water; (c) gas heated in a rigid sealed container; (d) a gas expanding inside a balloon as it is heated at atmospheric pressure.
 11. A gas is taken from state A ($P = 1 \text{ atm}$, $V = 2 \text{ L}$, $T = 300 \text{ K}$) to state B ($P = 2 \text{ atm}$, $V = 1 \text{ L}$, $T = 300 \text{ K}$) via two different paths. Path 1: isothermal compression. Path 2: isochoric heating then isobaric compression. (a) What is ΔU for each path? (b) Are Q and W the same for each path? Explain.
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First Law Problems

12. A system absorbs 800 J of heat and does 300 J of work on its surroundings. Find ΔU .
 13. A system has 600 J of work done on it and releases 200 J of heat. Find ΔU .
 14. In an isochoric process, 400 J of heat is removed from a gas. Find ΔU and W .
 15. A gas expands at constant pressure $P = 150 \text{ kPa}$ from $V_1 = 0.020 \text{ m}^3$ to $V_2 = 0.035 \text{ m}^3$ while absorbing 500 J of heat. Find W and ΔU .
 16. An adiabatic process compresses a gas, doing 250 J of work on it. Find ΔU and the sign of the temperature change.
 17. Over one complete thermodynamic cycle: the system absorbs 1200 J of heat from a hot reservoir and 800 J of heat is rejected to a cold reservoir. Find W (net work done by the system) and verify $\Delta U_{\text{cycle}} = 0$.
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Entropy Concept Exercises

18. Rank the following in order of increasing entropy: (a) 1 kg of ice at 0°C , (b) 1 kg of liquid water at 20°C , (c) 1 kg of steam at 120°C , (d) 1 kg of liquid water at 80°C .
 19. Does entropy increase, decrease, or remain the same in each process? (a) Water freezing at 0°C ; (b) a gas expanding into a vacuum; (c) heat flowing reversibly from T_H to T_C ; (d) a ball bearing rolling to a stop due to friction; (e) electricity flowing through a resistor.
 20. A reversible process transfers 1000 J of heat from a reservoir at 500 K to a system at 300 K. Find ΔS for the reservoir, the system, and the universe.
 21. An irreversible process transfers 1000 J of heat from a reservoir at 500 K directly to a cold body at 200 K. Find ΔS for each and for the universe. Compare to the result in Problem 3 and explain the difference.
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Applied Thermodynamic Reasoning

22. A power plant burns coal to produce steam at $T_H = 600^{\circ}\text{C}$ (873 K) and exhausts to a river at $T_C = 20^{\circ}\text{C}$ (293 K). (a) Find the Carnot (maximum possible) efficiency. (b) The plant actually operates at 35% efficiency. By how much does this fall short of the Carnot limit? (c) If the plant produces 1.0 GW of electrical output, how much heat must it reject to the river per second?
23. A living cell takes in chemical energy (from glucose) and produces work (muscle contraction) and heat. (a) Which thermodynamic law governs the total energy budget? (b) Which law explains why the cell produces heat even while doing useful work? (c) Does the cell's operation violate the Second Law?
24. Explain why a perfect insulator (a material that prevents all heat flow) would also eliminate convection through a wall but not radiation. What modification would be needed to also eliminate radiation?
25. Design an ideal experiment to test whether a proposed new engine violates the Second Law. What measurements would you take, and what result would indicate a violation?

Chapter 33 Checkpoint

Before moving on to Chapter 34, confirm you can do each of the following.

I can...	See Section
Classify systems as open, closed, or isolated	33.2
Distinguish state variables from path functions	33.3
State and explain the Zeroth Law; explain its role in defining temperature	33.4
State the First Law $\Delta U = Q - W$; apply correct sign conventions	33.5
Apply the First Law to isothermal, adiabatic, isochoric, and isobaric processes	33.7
State the Second Law in both Clausius and Kelvin-Planck formulations	33.8
Explain entropy as microscopic disorder; state $S = k_B \ln \Omega$	33.9
Explain why heat spontaneously flows from hot to cold using entropy reasoning	33.10
State the Third Law; explain why 0 K cannot be reached	33.11
Calculate Carnot efficiency from reservoir temperatures	33.12
Identify and correct common misconceptions about thermodynamic laws	33.13

Continue to Chapter 34: Heat Engines, Entropy, and Efficiency →